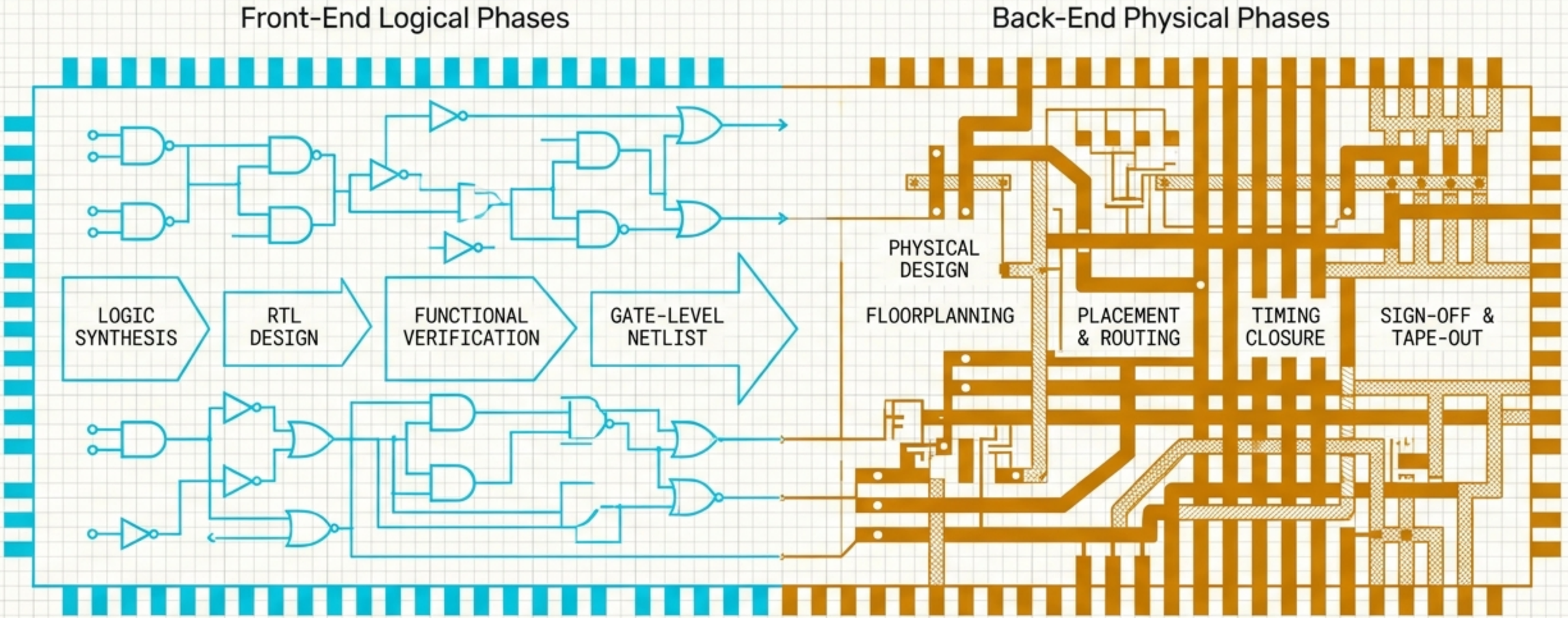
Blueprint to Silicon

Demystifying the Application-Specific Integrated Circuit (ASIC) Design Flow



FROM LOGIC GATES TO PHYSICAL GEOMETRY

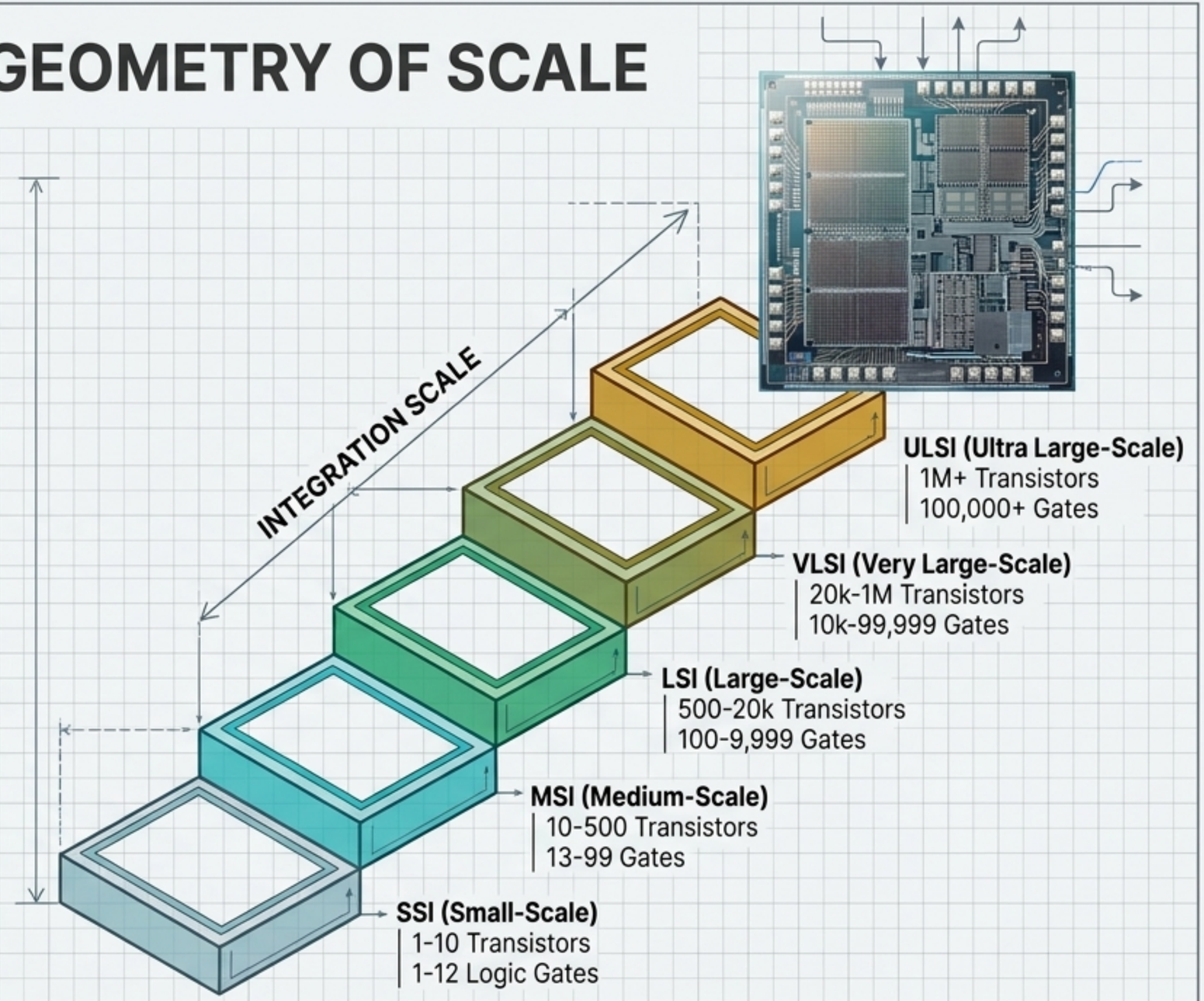
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THE GEOMETRY OF SCALE

- An Integrated Circuit (IC) is a semiconductor wafer holding millions of tiny electronic components within a single die substrate.

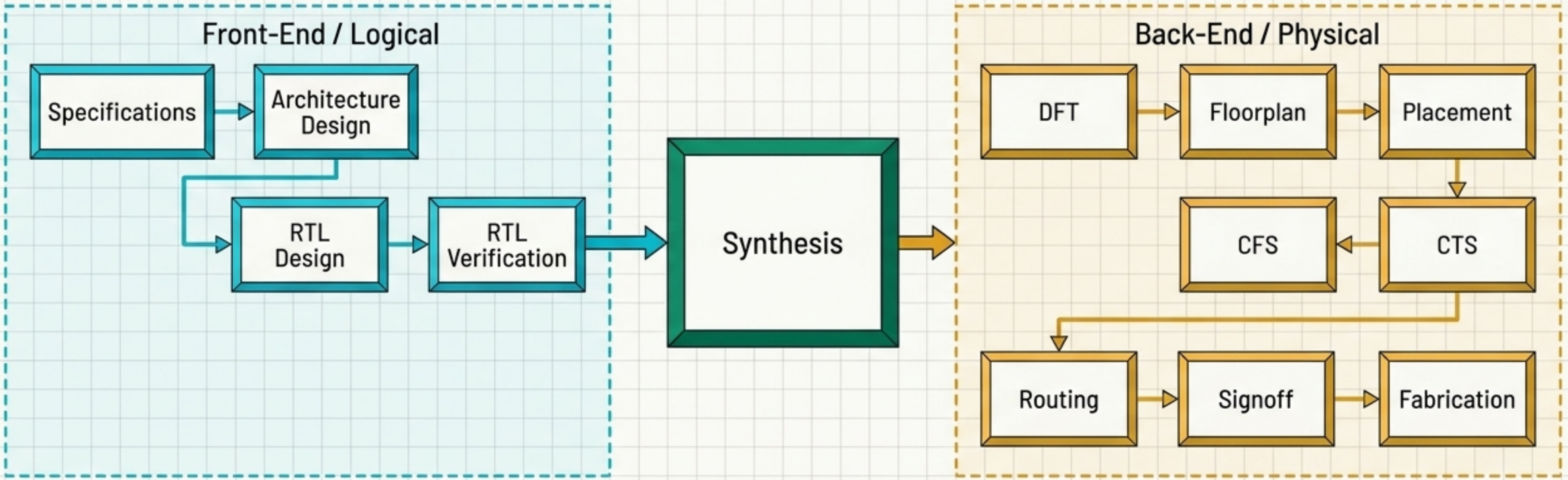
→ Designed using **Electronic Design Automation** (EDA) **tools** to optimize Area, Power, and Timing.

→ The final blueprint is sent to a semiconductor foundry as a `GDSII` file for fabrication.



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THE TWO HEMISPHERES OF CHIP DESIGN



Abstract, human-readable logic.
Technology independent.

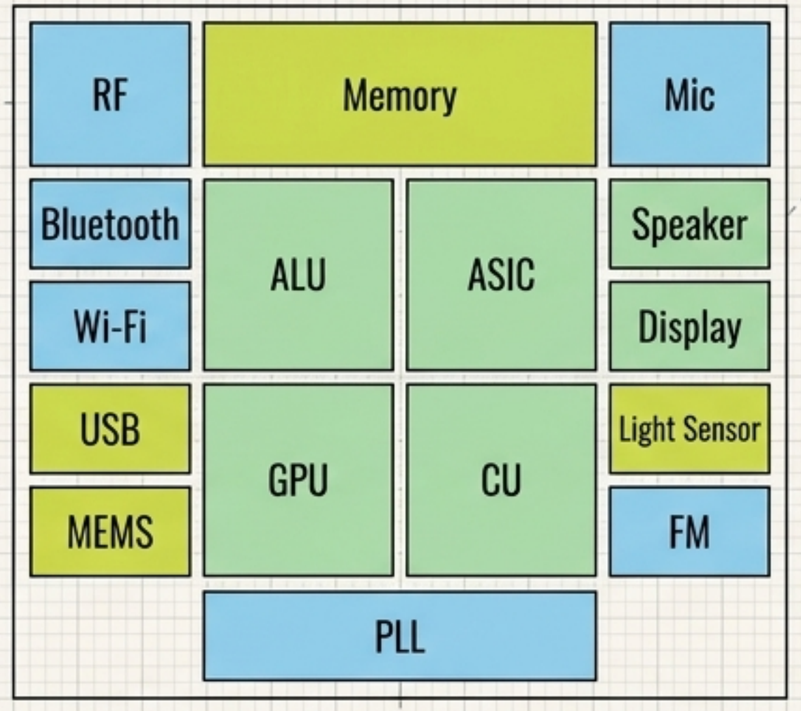
Physical layout and geometry.
Manufacturable and technology-dependent.

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ESTABLISHING THE ARCHITECTURAL FOUNDATION

Specifications

Marketing and product teams provide expected features and customer requirements.



Architecture

Engineers translate specs into a structural block diagram defining connected processors, memories, and interfaces.

The Output

Early estimations of block area, required power, and total design cost are established here.

- Digital (Light Green)
- Analog/RF (Light Blue)
- Mixed Signal (Yellow-Green)

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WRITING THE LOGIC AT THE REGISTER TRANSFER LEVEL

RTL DESIGN

- Uses Hardware Description Languages (Verilog or VHDL).
- Describes exactly how data is mathematically transformed as it passes from register to register.
- Uses abstract combinational and sequential circuit descriptions.

```
module counter(
  input clk, rstn, load,
  input [1:0] in,

  output reg [1:0] out);

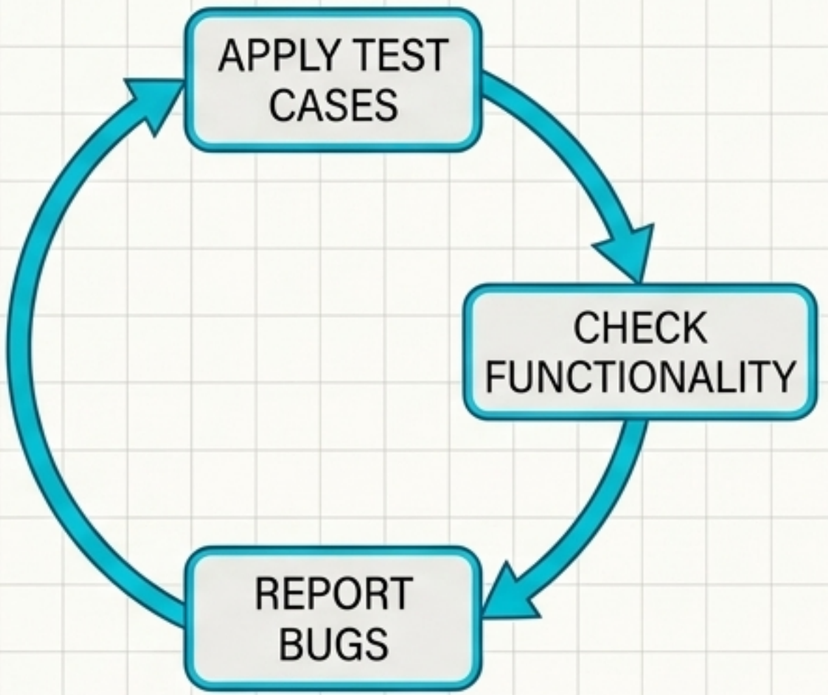
  always @(posedge clk)

    if (!rstn) out <= 2'b0;
    else if (load) out <= in;
    else out <= out + 1;

endmodule
```

RTL code for Counter

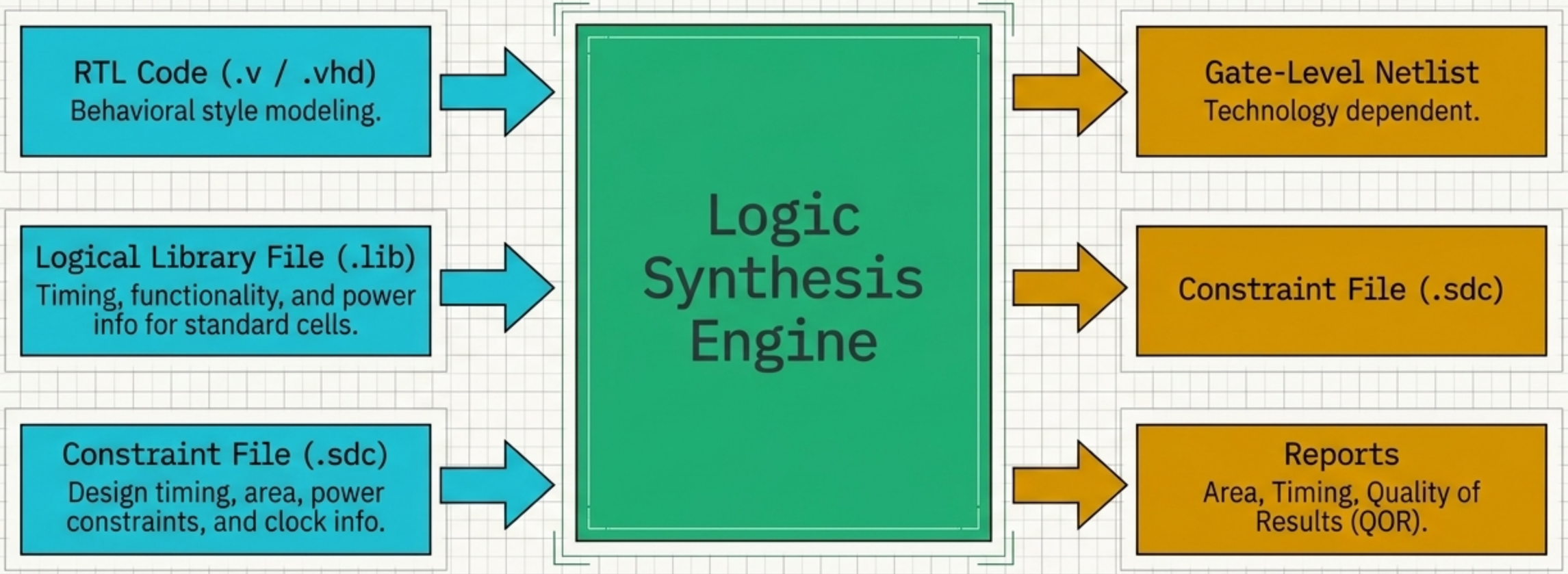
RTL VERIFICATION



Functional verification of the design. Mistakes found here send the code directly back to the RTL design team.

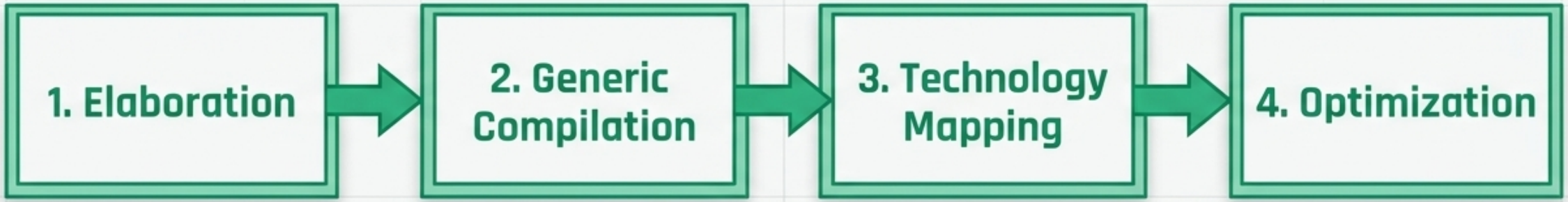
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The Synthesis Engine



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Inside the Synthesis Flow



Expanding HDL descriptions to represent all module instances into unique objects.

Converting RTL into simple logic gates (creating a technology-independent schematic).

Mapping generic gates to actual, physical technology-dependent logic gates available in the te .lib libraries.

Restructuring the mapped netlist to ensure it meets strict Area, Power, and Timing constraints.

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The Code Transformation

Before: RTL Verilog

```
always @(posedge clk)
  if (!rstn) out <= 2'b0;
  else out <= out + 1;
```

Abstract behavioral logic (if/else). Technology independent.

After: Gate-Level Netlist

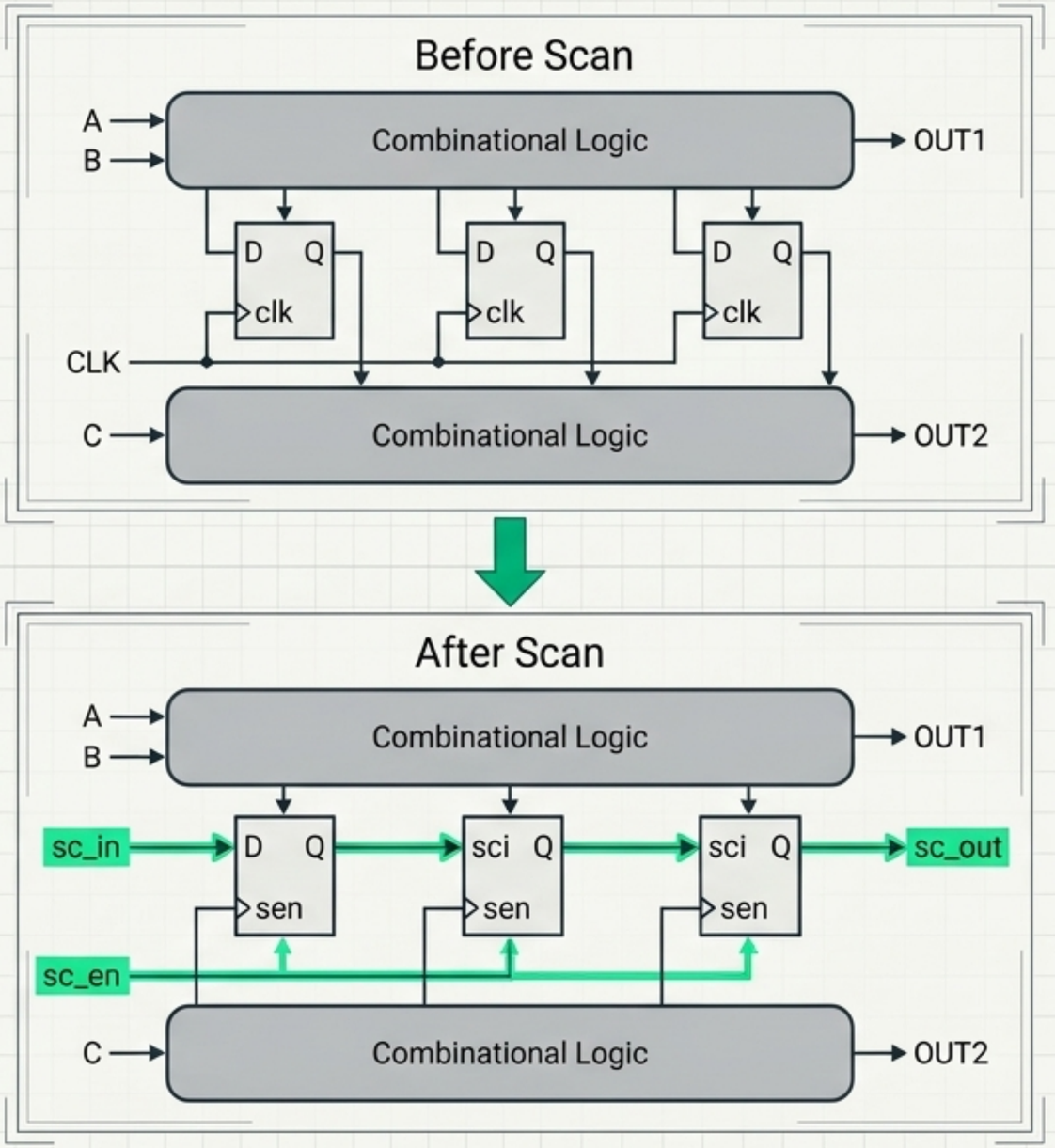
```
FFPQ1 out_reg_1 (.D(N7),
.CK(clk), .Q(out[1]));
NAN2D1 UB (.A1(out[0]),
.A2(n5), .Z(n8));
INVD1 U10 (.A(load), .Z(n));
```

Explicit physical logic gates (NAND, INVD). Technology dependent.

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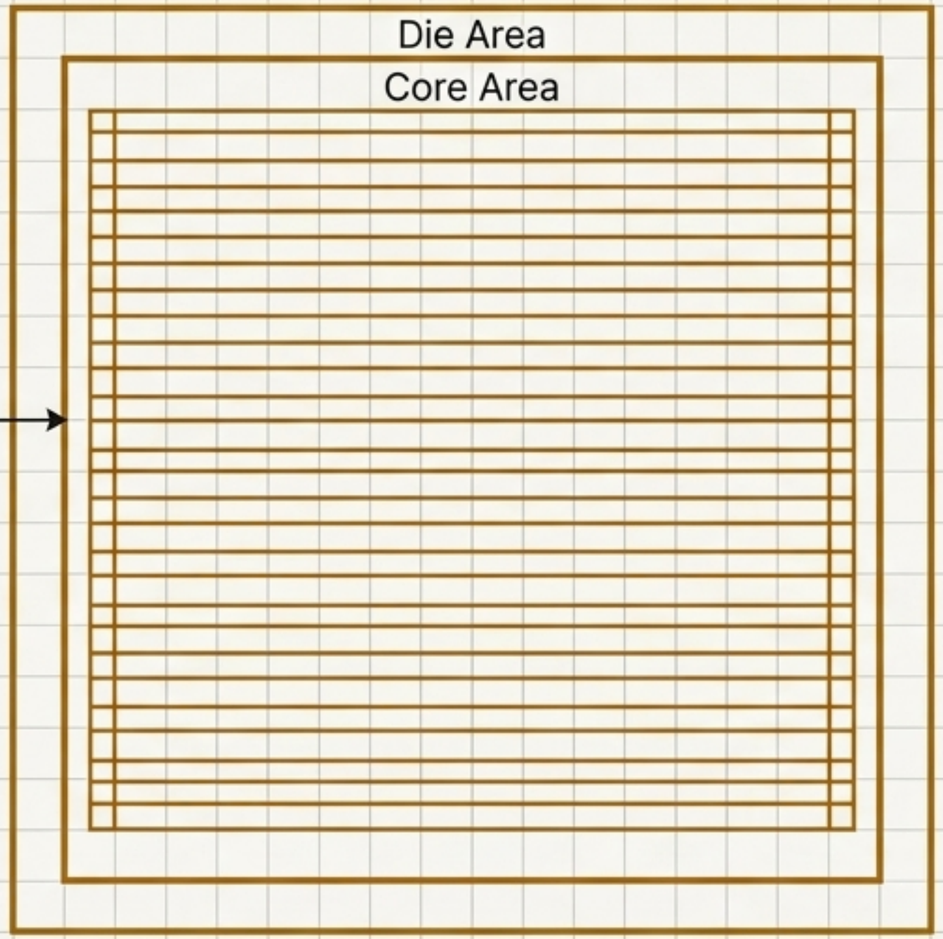
Design for Testability (DFT)

- DFT facilitates post-production testing.
- Extra logic (Scan Chains) is inserted alongside functional design logic without disrupting standard operations.
- This stage generates an ATPG (Automatic Test Pattern Generator) file.



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Floorplanning the Chip Anatomy



Site Rows
Like empty shelves waiting for standard cell books

$$\text{Aspect Ratio} = \frac{\text{Height}}{\text{Width}}$$

(Determines the shape of the block)

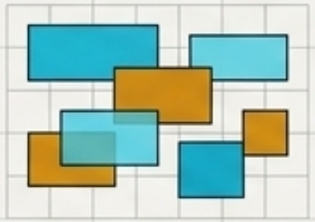
$$\text{Core Utilization} = \frac{(\text{Standard Cells Area} + \text{Macros Area})}{\text{Total Core Area}}$$

Additional Steps:
Port/pad placement, macro placement, and power planning occur here.

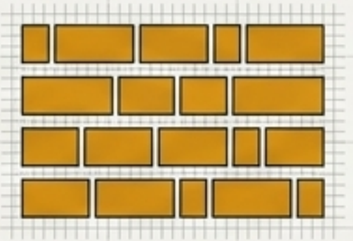
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The Physical Phal Phasing: Global vs. Detailed

Placement (Standard Cells)



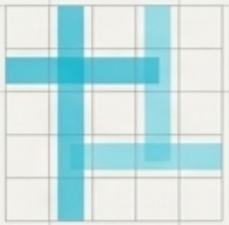
Rough, approximate assignment of cells inside the core area. Cells may overlap. Used to check initial congestion and timing.



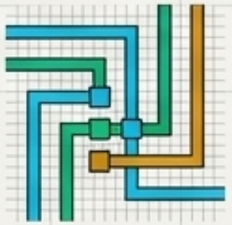
Exact, "legalized" assignment. Cells are snapped perfectly into the site rows with zero overlapping.

Routing (Metal Layers)

Plans the approximate paths for wires across coarse routing grids.

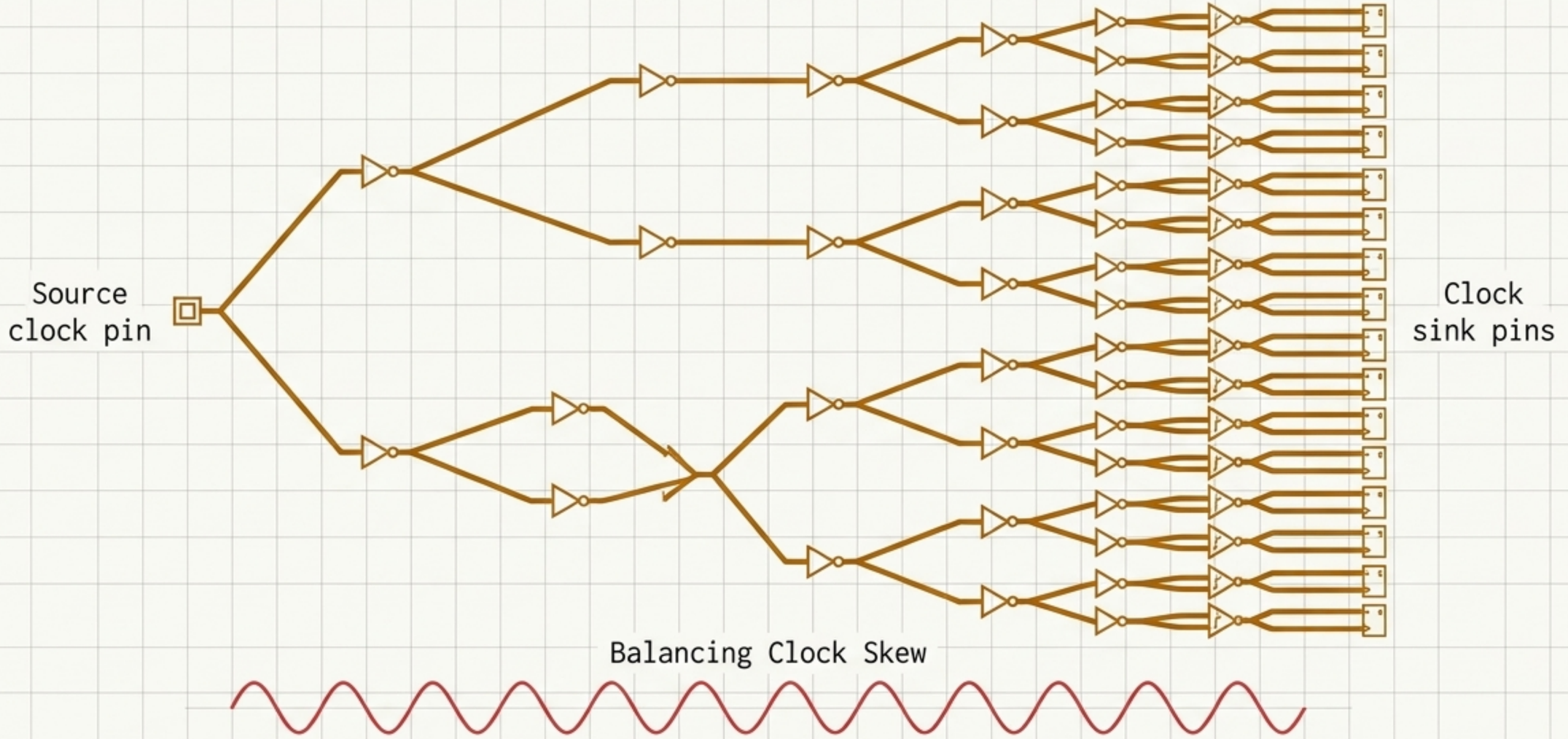


Lays down the exact, physical metal layer geometries and vias, ensuring all Design Rule Checks (DRCs) are met.



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Clock Tree Synthesis (CTS)



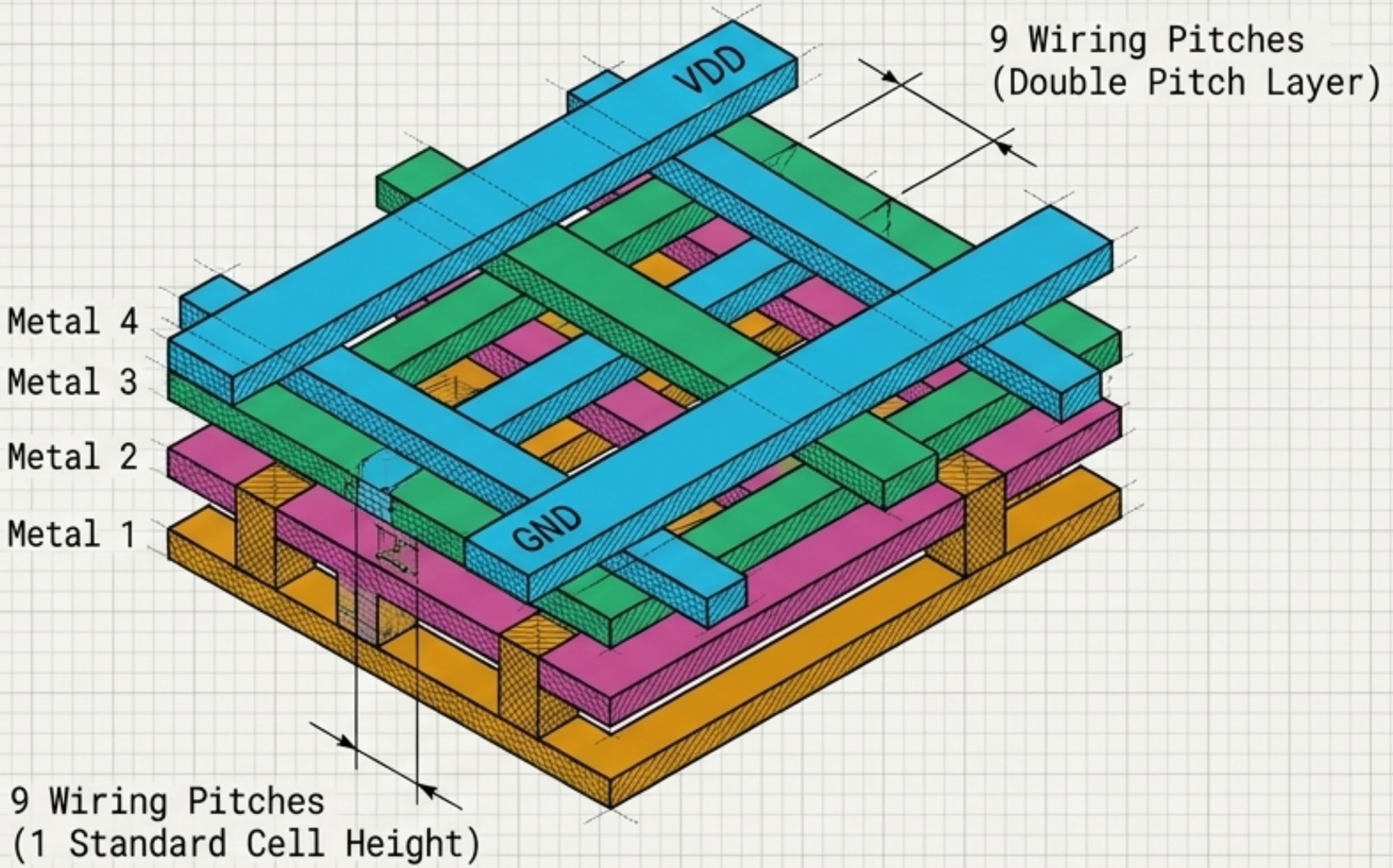
CTS builds the clock tree to distribute the clock signal perfectly across the chip.

Primary goals: Balance clock skew (arrival time differences) and minimize insertion delay, ensuring power and timing constraints are met.

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Routing the Physical Connections

- Before this stage, connections between macros, cells, and ports are purely logical mathematical lines.
- Routing converts these logical connections into physical metal tracks across multiple layered grids.



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Final Signoff to Silicon

Timing Signoff

SPEF (Standard Parasitic Exchange Format) extraction is performed.

Final timing is fixed using **Tempus/PrimeTime**.

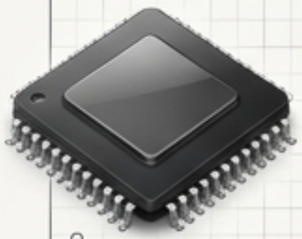
Physical Signoff

Rigorous quality and performance checks before tape-out.
Includes **DRC** (Design Rule Check), **LVS** (Layout vs. Schematic), and **LEC** (Logic Equivalence Check).

The finalized design is converted into the **GDSII** file.

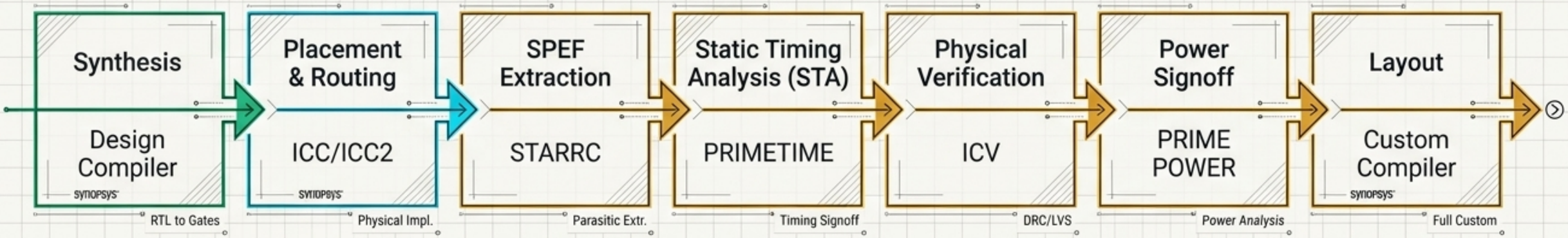
Fabrication

The GDSII blueprint is sent to the foundry where manufacturing processes transform the digital files into a physical silicon die.



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The EDA Toolchain Map



The ASIC design flow transforms abstract intent into precise physical geometry through a rigid sequence of interdependent software engines.

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